ZOLEDRONIC ACID

Once yearly (osteoporosis)

Acute-phase reaction

CHEMICAL NAME [1-hydroxy-2-(1H-imidazol-1-yl)ethane-1,1-diyl]bis(phosphonic acid) (monohydrate)

CLASS Intravenous bisphosphonate

TRADE NAMES Aclasta / Reclast (osteoporosis) · Zometa (oncology) · Zoledronate

THERAPEUTIC USES / INDICATIONS

- Osteoporosis (yearly infusion) incl. after hip fracture.
- Hypercalcaemia of malignancy.
- Prevention of skeletal events in bone metastases and multiple myeloma.
- Paget's disease.

ADMINISTRATION

- **IV infusion over at least 15 min** in a separate line; hydrate well before.
- Not with calcium-containing fluids (Ringer's).
- Check calcium, vitamin D and renal function before each dose.

DOSAGES

- **Osteoporosis:** 5 mg IV once yearly (usually for 3 years).
- **Hypercalcaemia of malignancy:** 4 mg IV single dose.
- **Bone metastases/myeloma:** 4 mg every 3–4 weeks (renally adjusted).
- **Paget's:** 5 mg single dose.

PHARMACOKINETICS

- Binds bone avidly; not metabolised.
- Unbound drug excreted unchanged in urine. Terminal **t½:** ~146 h (bone years).

PHARMACODYNAMICS

- Potent inhibition of **osteoclast-mediated bone resorption** → ↑ bone density, ↓ serum calcium.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Acute-phase reaction (fever, flu-like symptoms, myalgia, arthralgia, headache) in first 3 days; mainly first infusion.
- Nausea.
- Eye inflammation (uveitis, conjunctivitis).

ADVERSE EFFECTS (SERIOUS)

- Hypocalcaemia.
- Renal impairment / AKI.
- Osteonecrosis of the jaw.
- Atypical femoral fractures; atrial fibrillation.

CONTRAINDICATIONS & CAUTIONS

- Hypocalcaemia.
- CrCl <35 mL/min (osteoporosis dose).
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- **Aminoglycosides, loop diuretics:** ↑ hypocalcaemia.
- **Nephrotoxic drugs (NSAIDs):** ↑ renal risk.

NURSING MANAGEMENT

- Check creatinine, calcium, vitamin D before infusion.
- Hydrate (at least 500 mL fluid) before.
- Paracetamol can reduce acute-phase reaction.
- Dental review before starting.
- Monitor for hypocalcaemia (tingling, cramps).

PATIENT EDUCATION

- Flu-like symptoms for a few days after the first infusion are common; paracetamol helps.
- Drink plenty of fluids before and after.
- Keep good dental hygiene; report jaw pain.
- Take calcium and vitamin D as prescribed.

KEY POINTS

- IV bisphosphonate — 5 mg yearly for osteoporosis.
- Acute-phase reaction; hypocalcaemia; renal toxicity.
- Infuse over ≥15 min; hydrate.
- Hypercalcaemia of malignancy 4 mg.